FENG, RUOCHEN

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EDUCATION

Columbia University — **New York City, US**
Master of Science in Data Science, GPA:3.83 / 4.00 — *Aug 2025 – Expected Dec 2026*
- Coursework: Gen AI using LLMs, Agentic AI, Causal Inference, Machine Learning, Applied Deep Learning

University of Toronto — **Toronto, CA**
Honours Bachelor of Science (H.B.Sc) with High Distinction, GPA:3.94 / 4.00 — *Sep 2021 – Jun 2025*
- Program: Mathematics and Statistics Double Major, Computer Science Minor
- Scholarship & Honors: C.L. Burton Scholarship, University College Alumni Scholarship, Dean's List Scholar

WORK EXPERIENCE

Google Inc. — *June 2026 – Sep 2026*
Incoming Data Scientist Intern — *Greater Seattle Area, US*
- Incoming Intern on **Google Maps Navigation Trips**, selected to develop **a driver-based trip-demand forecasting model** using time-series feature engineering, regression/forecasting models, and baseline evaluation against Google's existing momentum-based methodology.
- Responsible for data assembly, model design, implementation, evaluation, and performance optimization, with prediction dashboards and **Gemini-powered** summaries to communicate forecast drivers, assumptions, and product-facing insights.

VeriSilicon Microelectronics Co., Ltd. — *Jul 2023 – Sep 2023*
Data Engineer Intern — *Shanghai, China*
- Built an automated financial reporting pipeline using **AWS S3/Lambda**, **Spark batch processing**, **Kafka ingestion**, ReportLab PDF generation, and SendGrid delivery, reducing recurring report generation time by ~50% and improving cross-team distribution reliability.
- Developed reusable **SQL/Python** data preparation workflows joining 100K+ experiment, customer, and usage records to support **A/B test readouts**, and **funnel analysis**; analysis informed product iterations that improved SMB adoption by 17.3%.

OCBC Bank Ltd. — *Apr 2023 – Jun 2023*
Risk Analyst Intern — *Shanghai, China*
- Consolidated and cleaned fragmented financial datasets (~10M+ transaction and position records) using **SQL** techniques, including **window functions**, **CTEs**, and joins across multiple schemas, to build unified exposure tables for downstream risk analysis.
- Designed an interactive **Tableau dashboard** for real-time portfolio risk monitoring, enabling faster decision-making and higher returns.
- Built and deployed **Python-based volatility forecasting models** (e.g., **GARCH**) to assess market dynamics and quantify interest rate impacts, improving prediction accuracy by **21.4%**, with version control managed via **Git**.

RESEARCH & PROJECTS

Deep Learning for Histopathology Image Analysis — *Sep 2025 – Present*
Machine Learning Researcher, Reya Lab, Columbia University Irving Medical Center
- Built supervised H&E tile-classification pipelines with an **ImageNet-pretrained WideResNet-50-2 backbone**, **PyTorch/scikit-learn** evaluation, class balancing, threshold tuning, and leave-one-slide-out validation across 43,213 annotated tiles and 5 tissue classes.
- Applied **StarDist 2D_versatile_he**, a CNN-based instance segmentation model, to high-resolution H&E tissue microarrays, producing 80K+ nuclear masks and an interactive HTML/JS viewer for spatial QA and model interpretability.

TravelMind: Multi-Agent LLM Travel Planning System — *Sep 2025 – Dec 2025*
Project Lead, Supervisor: Adj. Prof. Spencer Luo
- Designed a multi-agent LLM travel-planning system with **React/Next.js frontend** and **Flask API architecture**, routing user requests through ConstraintParser, DestinationRecommender, ItineraryPlanner, DetailEnricher, and PlanEnhancer agents.
- Specified dual-model enhancement using **OpenAI/GPT-4o-mini** and **a Hugging Face travel model**, with REST endpoints, progressive enrichment, fallback handling, and demo flows for destination discovery and existing-plan improvement.

TECHNICAL SKILLS

- Programming Languages: Python, Java, R, SQL, MATLAB, C, C++, LaTeX, SAS, Stata
- Tools: Spark, Kafka, AWS (S3, EC2), GCP (Vertex AI), Git, Jupyter, MySQL, PostgreSQL, Tableau, Power BI, Matplotlib
- Libraries: Numpy, Pandas, PyTorch, Scikit-learn, TensorFlow, Matplotlib, RStan, tidyverse, StarDist